

Overview of PHP

PHP ek **server-side scripting language** hai jo dynamic websites aur web applications banane ke liye use hoti hai. PHP ka full form **Hypertext Preprocessor** hai. Ye server par run hoti hai aur browser ko final HTML output bhejti hai. PHP ko HTML ke saath easily embed kiya ja sakta hai. Iska use login system, forms, e-commerce websites, dashboards, blogs, APIs etc. banane me hota hai. PHP open-source hai, free hai aur beginner-friendly language mani jati hai. Ye MySQL database ke saath commonly use hoti hai. WordPress jaise popular platforms bhi PHP par based hain.

Example 1: Simple PHP Program

```
<?php
echo "Hello World";
?>
```

Output

Hello World

Example 2: PHP with HTML

```
<!DOCTYPE html>
<html>
<body>

<h1>My First PHP Page</h1>

<?php
echo "Welcome to PHP";
?>

</body>
</html>
```

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Features of PHP

PHP me bahut useful features hote hain jo web development ko easy banate hain. Ye open-source language hai, isliye free me use ki ja sakti hai. PHP platform independent hai, matlab Windows, Linux aur Mac sab par chalti hai. Ye

HTML ke saath directly embed ho sakti hai. PHP databases jaise MySQL ko support karti hai jisse data store aur retrieve karna easy hota hai. PHP fast execution provide karti hai aur large community support bhi available hai. Isme built-in functions ka huge collection hota hai jo development speed badhata hai. PHP cookies, sessions aur file handling bhi support karti hai.

Example 1: PHP Embedded in HTML

```
<h2>User Information</h2>

<?php
echo "Name: Adesh";
?>
```

Example 2: Database Connection Syntax

```
<?php
$conn = mysqli_connect("localhost", "root", "", "testdb");

if($conn){
    echo "Connected Successfully";
}
?>
```

Example 3: Using Built-in Function

```
<?php
echo strlen("PHP Language");
?>
```

Output

12

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Benefits and Applications of PHP

PHP web development ke liye ek powerful aur easy language hai. Iska sabse bada benefit ye hai ki ye free aur open-source hai. PHP beginner-friendly hai aur easily HTML ke saath work karti hai. Ye fast execution aur database connectivity provide karti hai. PHP ka use dynamic websites aur web applications banane me hota hai. Isse login

systems, online shopping websites, CMS, APIs, dashboards aur form handling applications banaye jate hain. PHP hosting support almost har server par available hota hai, isliye deployment bhi easy hota hai.

Applications of PHP

- E-commerce websites
- Blog websites
- Student management systems
- Login and registration systems
- Content Management Systems (CMS)

Example 1: Login Message

```
<?php
$username = "Admin";

echo "Welcome " . $username;
?>
```

Output

Welcome Admin

Example 2: Simple Calculator

```
<?php
$a = 10;
$b = 5;

echo $a + $b;
?>
```

Output

15

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PHP Installation (XAMPP, WAMP)

PHP run karne ke liye local server environment ki zarurat hoti hai. Iske liye mostly **XAMPP** aur **WAMP** use kiye jate hain. XAMPP me Apache Server, MySQL Database aur PHP already included hote hain. WAMP bhi same kaam karta hai lekin mainly Windows ke liye hota hai. Installation ke baad PHP files `htdocs` folder me rakhi jati hain aur browser me `localhost` ke through run ki jati hain. XAMPP beginner-friendly aur easy setup provide karta hai.

XAMPP Installation Steps

1. XAMPP download karo
2. Install karo
3. XAMPP Control Panel open karo
4. Apache start karo
5. `htdocs` folder me PHP file save karo
6. Browser me run karo

Example Folder Path

```
C:\xampp\htdocs\test.php
```

Example PHP File

```
<?php
echo "PHP Installed Successfully";
?>
```

Run in Browser

```
http://localhost/test.php
```

Output

```
PHP Installed Successfully
```

```
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```

Basic Syntax of PHP

PHP code hamesha `<?php ?>` tags ke andar likha jata hai. Har statement ke end me semicolon `;` lagana important hota hai. Output display karne ke liye `echo` ya `print` use kiya jata hai. PHP case-sensitive language hai, matlab variable names uppercase aur lowercase me different hote hain. Comments likhne ke liye `//` single-line aur `/* */` multi-line comments use hote hain. PHP code HTML ke andar bhi likha ja sakta hai.

Syntax Structure

```
<?php
// PHP code here
?>
```

Example 1: Print Output

```
<?php
echo "Welcome to PHP";
?>
```

Output

```
Welcome to PHP
```

Example 2: Using Variables

```
<?php
$name = "Adesh";

echo $name;
?>
```

Output

```
Adesh
```

Example 3: Comments

```
<?php
// Single-line comment
```

```
/*  
Multi-line  
comment  
*/  
?>
```

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Variables in PHP

Variables data store karne ke liye use hote hain. PHP me variable hamesha `$` symbol se start hota hai. Variable ka naam letter ya underscore `_` se start hona chahiye. PHP dynamically typed language hai, isliye data type automatically detect ho jata hai. Variables me numbers, text, arrays ya other values store ki ja sakti hain. Variable names case-sensitive hote hain, matlab `$name` aur `$Name` different variables hain.

Variable Syntax

```
$variable_name = value;
```

Example 1: String Variable

```
<?php  
$name = "Adesh";  
  
echo $name;  
?>
```

Output

Adesh

Example 2: Number Variables

```
<?php  
$a = 10;  
$b = 20;  
  
echo $a + $b;  
?>
```

Output

30

Example 3: Multiple Variables

```
<?php
$city = "Nagpur";
$age = 22;

echo $city;
echo $age;
?>
```

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Data Types in PHP

Data type variable me store hone wale data ka type batata hai. PHP automatically variable ka data type identify kar leti hai. Common data types me String, Integer, Float, Boolean, Array aur NULL include hote hain. String text values store karta hai, Integer whole numbers, Float decimal numbers aur Boolean true/false values store karta hai. Arrays multiple values ko ek variable me store karte hain. Correct data type use karne se program efficient aur easy to manage banta hai.

Example 1: String

```
<?php
$name = "Adesh";

echo $name;
?>
```

Example 2: Integer and Float

```
<?php
$age = 22;
$price = 99.99;

echo $age;
echo $price;
?>
```

Example 3: Boolean

```
<?php
$isLogin = true;

var_dump($isLogin);
?>
```

Output

```
bool(true)
```

Example 4: Array

```
<?php
$colors = array("Red", "Blue", "Green");

print_r($colors);
?>
```

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Constants in PHP

Constants aise values hoti hain jinhe program me ek baar define karne ke baad change nahi kiya ja sakta. Variables ki tarah constants ke aage `$` symbol nahi lagta. PHP me constants create karne ke liye `define()` function use hota hai. Constants mostly fixed values store karne ke liye use kiye jate hain jaise website name, PI value ya database configuration. Constants globally accessible hote hain, yani program ke kisi bhi part me use kiye ja sakte hain.

Syntax

```
define("CONSTANT_NAME", value);
```

Example 1: Simple Constant

```
<?php
define("SITE_NAME", "ICE Computers");

echo SITE_NAME;
?>
```


Output

ICE Computers

Example 2: Numeric Constant

```
<?php
define("PI", 3.14);

echo PI;
?>
```

Output

3.14

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Control Structures in PHP

Control structures program ke flow ko control karte hain. Inki help se conditions check ki ja sakti hain aur repeated tasks perform kiye ja sakte hain. PHP me mainly `if-else`, `switch` aur loops use hote hain. `if-else` conditions check karta hai, `switch` multiple choices handle karta hai aur loops repeated execution ke liye use hote hain. Ye structures decision making aur automation ke liye bahut important hote hain.

Example 1: If-Else

```
<?php
$age = 18;

if($age >= 18){
    echo "Eligible";
}
else{
    echo "Not Eligible";
}
?>
```

Example 2: Switch

```
<?php
$day = 2;

switch($day){
    case 1:
        echo "Monday";
        break;

    case 2:
        echo "Tuesday";
        break;
}
?>
```

Example 3: Loop

```
<?php
for($i=1; $i<=5; $i++){
    echo $i;
}
?>
```

Output

12345

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If-Else in PHP

if-else statement conditions check karne ke liye use hota hai. Agar condition true hoti hai toh **if** block execute hota hai, warna **else** block execute hota hai. Ye decision making ke liye bahut useful hota hai jaise login validation, marks checking aur eligibility systems me. Multiple conditions check karne ke liye **elseif** bhi use kiya ja sakta hai.

Syntax

```
if(condition){
    // code
}
else{
    // code
}
```

Example 1: Simple If-Else

```
<?php
$marks = 40;

if($marks >= 35){
    echo "Pass";
}
else{
    echo "Fail";
}
?>
```

Output

Pass

Example 2: Using Elseif

```
<?php
$marks = 85;

if($marks >= 75){
    echo "Distinction";
}
elseif($marks >= 35){
    echo "Pass";
}
else{
    echo "Fail";
}
?>
```

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Switch in PHP

switch statement multiple conditions check karne ke liye use hota hai. Jab ek variable ki value ko different cases se compare karna ho tab switch use karna better hota hai. Har case ke baad **break** likha jata hai taaki execution wahi stop ho jaye. Agar koi case match nahi hota toh **default** block execute hota hai. Ye menu systems, day selection aur option handling me useful hota hai.

Syntax

```
switch(variable){  
    case value1:  
        // code  
        break;  
  
    case value2:  
        // code  
        break;  
  
    default:  
        // code  
}
```

Example 1: Day Example

```
<?php  
$day = 3;  
  
switch($day){  
    case 1:  
        echo "Monday";  
        break;  
  
    case 2:  
        echo "Tuesday";  
        break;  
  
    case 3:  
        echo "Wednesday";  
        break;  
  
    default:  
        echo "Invalid Day";  
}  
?>
```

Output

wednesday

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Loops in PHP

Loops repeated execution ke liye use hote hain. Jab same code ko multiple times run karna ho tab loops helpful hote hain. PHP me mainly `for`, `while`, `do-while` aur `foreach` loops use hote hain. `for` loop fixed repetitions ke liye,

`while` condition-based execution ke liye aur `foreach` arrays ke elements access karne ke liye use hota hai. Loops programming me automation aur data processing ko easy banate hain.

Example 1: For Loop

```
<?php
for($i = 1; $i <= 5; $i++){
    echo $i . "<br>";
}
?>
```

Output

```
1
2
3
4
5
```

Example 2: While Loop

```
<?php
$num = 1;

while($num <= 3){
    echo $num;
    $num++;
}
?>
```

Output

```
123
```

Example 3: Foreach Loop

```
<?php
$colors = array("Red", "Blue", "Green");

foreach($colors as $color){
    echo $color . "<br>";
}
```

```
}  
?>
```

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PHP with HTML (Embedding PHP)

PHP ko HTML ke andar directly embed kiya ja sakta hai. Isse dynamic webpages banana easy ho jata hai. PHP code `<?php ?>` tags ke andar likha jata hai aur HTML ke saath mix karke use kiya jata hai. Browser ko final output HTML format me milta hai. PHP embedding ka use dynamic content show karne, user data display karne aur forms handle karne ke liye hota hai.

Syntax

```
<?php  
// PHP code  
?>
```

Example 1: PHP Inside HTML

```
<!DOCTYPE html>  
<html>  
<body>  
  
<h1>Welcome Page</h1>  
  
<?php  
echo "Hello Adesh";  
?>  
  
</body>  
</html>
```

Output

```
Welcome Page  
Hello Adesh
```

Example 2: Display Variable in HTML

```
<?php
$name = "Adesh";
?>

<html>
<body>

<h2><?php echo $name; ?></h2>

</body>
</html>
```

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Functions in PHP (Creation and Usage)

Functions reusable code blocks hote hain jo specific task perform karte hain. Function ko ek baar define karke multiple times use kiya ja sakta hai. Isse code reusable aur organized banta hai. PHP me function create karne ke liye `function` keyword use hota hai. Functions parameters receive kar sakte hain aur value return bhi kar sakte hain. Large programs me functions code management aur debugging ko easy banate hain.

Syntax

```
function functionName(){
    // code
}
```

Example 1: Simple Function

```
<?php
function greet(){
    echo "Welcome to PHP";
}

greet();
?>
```

Output

Welcome to PHP

Example 2: Function with Parameters

```
<?php
function add($a, $b){
    echo $a + $b;
}

add(10, 20);
?>
```

Output

30

Example 3: Function Returning Value

```
<?php
function square($num){
    return $num * $num;
}

echo square(5);
?>
```

Output

25

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Built-in Functions in PHP

PHP me bahut saare built-in functions already available hote hain jo common tasks ko easy banate hain. In functions ka use string handling, arrays, math operations, date-time aur file handling ke liye hota hai. Built-in functions use karne se coding fast aur simple ho jati hai. Har function ka specific purpose hota hai aur predefined syntax hota hai.

Example 1: String Length Function

```
<?php
$text = "Hello PHP";
```



```
echo strlen($text);  
?>
```

Output

9

Example 2: Maximum Number Function

```
<?php  
echo max(10, 50, 30);  
?>
```

Output

50

Example 3: Date Function

```
<?php  
echo date("d-m-Y");  
?>
```

Output

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Example 4: String Uppercase

```
<?php  
echo strtoupper("php language");  
?>
```

Output

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Arrays in PHP

Array ek special variable hota hai jo multiple values ko single variable me store karta hai. PHP me arrays data collection handle karne ke liye use hote hain. Arrays ka use student names, marks, products ya multiple records store karne me hota hai. PHP me mainly Indexed Array, Associative Array aur Multidimensional Array use hote hain. Array ke elements index ya key ke through access kiye jate hain.

Syntax

```
$array_name = array(value1, value2, value3);
```

Example 1: Indexed Array

```
<?php
$colors = array("Red", "Blue", "Green");

echo $colors[0];
?>
```

Output

```
Red
```

Example 2: Associative Array

```
<?php
$student = array(
    "name" => "Adesh",
    "age" => 22
);

echo $student["name"];
?>
```

Output

```
Adesh
```

Example 3: Loop Through Array

```
<?php
$nums = array(10, 20, 30);

foreach($nums as $num){
    echo $num . "<br>";
}
?>
```

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Types of Arrays in PHP

PHP me mainly 3 types ke arrays hote hain: Indexed Array, Associative Array aur Multidimensional Array. Indexed array me numeric indexes use hote hain. Associative array me custom keys use hoti hain. Multidimensional array me arrays ke andar arrays store kiye jate hain. Arrays large amount of related data ko easily manage karne me help karte hain.

1. Indexed Array

Numeric indexes automatically assign hote hain.

```
<?php
$fruits = array("Apple", "Mango", "Orange");

echo $fruits[1];
?>
```

Output

Mango

2. Associative Array

Custom keys use hoti hain.

```
<?php
$student = array(
    "name" => "Adesh",
    "age" => 22
);
```

```
echo $student["age"];  
?>
```

Output

22

3. Multidimensional Array

Array ke andar multiple arrays hote hain.

```
<?php  
$marks = array(  
    array("Adesh", 80),  
    array("Rahul", 75)  
);  
  
echo $marks[0][0];  
?>
```

Output

Adesh

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Array Functions in PHP

PHP me arrays ko manage karne ke liye bahut saare built-in array functions available hote hain. Inka use counting, adding, removing, searching aur processing ke liye hota hai. Array functions coding ko easy aur fast bana dete hain. Common functions me `count()`, `array_push()`, `array_pop()` aur `in_array()` include hote hain.

Example 1: count()

Array me total elements count karta hai.

```
<?php  
$colors = array("Red", "Blue", "Green");  
  
echo count($colors);  
?>
```

Output

```
3
```

Example 2: array_push()

Array ke end me new element add karta hai.

```
<?php
$nums = array(10, 20);

array_push($nums, 30);

print_r($nums);
?>
```

Example 3: array_pop()

Last element remove karta hai.

```
<?php
$fruits = array("Apple", "Mango", "Orange");

array_pop($fruits);

print_r($fruits);
?>
```

Example 4: in_array()

Check karta hai value array me present hai ya nahi.

```
<?php
$colors = array("Red", "Blue");

if(in_array("Red", $colors)){
    echo "Found";
}
?>
```

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Sorting Arrays in PHP

Sorting ka use array elements ko ascending ya descending order me arrange karne ke liye hota hai. PHP me different sorting functions available hain jaise `sort()`, `rsort()`, `asort()` aur `ksort()`. `sort()` ascending order me sort karta hai aur `rsort()` descending order me. Associative arrays ke liye `asort()` values ke basis par aur `ksort()` keys ke basis par sorting karta hai.

Example 1: sort() – Ascending Order

```
<?php
$nums = array(40, 10, 30, 20);

sort($nums);

print_r($nums);
?>
```

Output

```
Array ( [0] => 10 [1] => 20 [2] => 30 [3] => 40 )
```

Example 2: rsort() – Descending Order

```
<?php
$nums = array(10, 50, 20);

rsort($nums);

print_r($nums);
?>
```

Example 3: asort() – Associative Array Sort by Value

```
<?php
$marks = array("A" => 80, "B" => 60);

asort($marks);

print_r($marks);
?>
```

Filtering Arrays in PHP

Filtering ka use array se specific elements select ya remove karne ke liye hota hai. PHP me `array_filter()` function commonly use hota hai. Ye function condition ke basis par elements filter karta hai. Filtering data cleaning, searching aur validation tasks me useful hoti hai. Custom function ke through filtering rules define kiye ja sakte hain.

Syntax

```
array_filter(array, function);
```

Example 1: Filter Even Numbers

```
<?php
$num = array(1, 2, 3, 4, 5, 6);

$result = array_filter($num, function($num){
    return $num % 2 == 0;
});

print_r($result);
?>
```

Output

```
Array ( [1] => 2 [3] => 4 [5] => 6 )
```

Example 2: Filter String Length

```
<?php
$names = array("Ram", "Adesh", "Aman");

$result = array_filter($names, function($name){
    return strlen($name) > 3;
});

print_r($result);
?>
```

Output

```
Array ( [1] => Adesh [2] => Aman )
```

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String Manipulation in PHP

String manipulation ka use text data ko modify aur process karne ke liye hota hai. PHP me strings handle karne ke liye bahut saare built-in functions available hain. In functions ka use string length check karne, uppercase/lowercase conversion, joining, splitting aur replacing ke liye hota hai. Ye functions form data processing aur text formatting me useful hote hain.

Example 1: String Length

```
<?php
$text = "Hello PHP";

echo strlen($text);
?>
```

Output

9

Example 2: Uppercase Conversion

```
<?php
echo strtoupper("php language");
?>
```

Output

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Example 3: Replace Text

```
<?php
$text = "I like Java";
```



```
echo str_replace("Java", "PHP", $text);  
?>
```

Output

```
I like PHP
```

Example 4: Join Strings

```
<?php  
$fname = "Adesh";  
$lname = "Lokhande";  
  
echo $fname . " " . $lname;  
?>
```

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Regular Expressions in PHP

Regular Expressions (Regex) ka use patterns match karne ke liye hota hai. PHP me regex mainly validation, searching aur text processing ke liye use hota hai. Iska use email validation, mobile number checking aur password verification me hota hai. PHP me regex ke liye `preg_match()` function commonly use kiya jata hai. Agar pattern match hota hai toh function `1` return karta hai, warna `0`.

Syntax

```
preg_match(pattern, string);
```

Example 1: Check Digits

```
<?php  
$text = "12345";  
  
if(preg_match("/^[0-9]+$/", $text)){  
    echo "Valid Number";  
}  
?>
```

Output

Valid Number

Example 2: Email Validation

```
<?php
$email = "test@gmail.com";

if(preg_match("/^[a-z0-9._]+@[a-z]+\.[a-z]{2,3}$/", $email)){
    echo "Valid Email";
}
?>
```

Output

Valid Email

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Working with Date and Time in PHP

PHP me date aur time handle karne ke liye built-in functions available hain. Inka use current date show karne, time display karne aur custom date formats banane ke liye hota hai. `date()` function sabse commonly use hota hai.

Date-time functions attendance systems, booking apps aur reports me useful hote hain. Different format characters use karke customized output generate kiya ja sakta hai.

Syntax

```
date(format);
```

Example 1: Current Date

```
<?php
echo date("d-m-Y");
?>
```

Output

07-05-2026

Example 2: Current Time

```
<?php
echo date("h:i:s A");
?>
```

Output

10:30:45 AM

Example 3: Full Date and Time

```
<?php
echo date("d-m-Y h:i:s A");
?>
```

Example Output

07-05-2026 10:30:45 AM

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